

Experiment 10: RSA

CODE:

```
import math
```

```
def gcd(a, h):
```

```
    temp = 0
```

```
    while(1):
```

```
        temp = a % h
```

```
        if (temp == 0):
```

```
            return h
```

```
        a = h
```

```
        h = temp
```

```
p = 3
```

```
q = 7
```

```
n = p*q
```

```
e = 2
```

```
phi = (p-1)*(q-1)
```

```
while (e < phi):
```

```
    # e must be co-prime to phi and
```

```
    # smaller than phi.
```

```
    if(gcd(e, phi) == 1):
```

```
        break
```

```
    else:
```

```
        e = e+1
```

```
# Private key (d stands for decrypt)
```

```
# choosing d such that it satisfies
```

```
#  $d * e = 1 + k * \text{totient}$ 
```

```
k = 2
```

```
d = (1 + (k*phi))/e
```

```
# Message to be encrypted
```

```
msg = 12.0
```

```
print("Message data = ", msg)
```

```
# Encryption  $c = (\text{msg}^e) \% n$ 
```

```
c = pow(msg, e)
```

```
c = math.fmod(c, n)
```

```
print("Encrypted data = ", c)
```

```
# Decryption  $m = (c^d) \% n$ 
```

```
m = pow(c, d)
```

```
m = math.fmod(m, n)
```

```
print("Original Message Sent = ", m)
```

OUTPUT:

Message data = 12.0

Encrypted data = 3.0

Original Message Sent = 12.0